School:		**School of Engineering & Technology**
Program:		**Computer Science & Engineering**
Branch:		**CSE**
1	Course Code	CSA303
2	Course Title	**Pattern Recognition**
3	Credits	3
4	Contact Hours (L-T-P)	3-0-0
	Course Status	CORE
5	Course Objective	**Students will try to learn t**o introduce the ideas of existing patterns and implement them based on data analysis. Also, to conceptualize the working of patterns explorations using computational algorithms. In addition to it, students will aim to become familiar with feature knowledge that can be extracted from available examples and generalize to form appropriate feature models.
6	Course Outcomes	On successful completion of this module students will be able to: CO 1. Define Pattern concept and random process ideas, understand mathematical background. CO 2. Explain preliminary models to understand the solution environment. CO 3. Apply of existing mathematical solutions to test problems. and Perform Subspace analysis for classification problems CO 4. Classify patterns using Bayesian Decision Theory. CO 5. Evaluate patterns using Parametric and Non-Parametric techniques. CO 6. Discuss trajectory of recent trend in pattern recognition & understand various real world applications.
7	Course Description	Pattern recognition theory and practice is concerned with the design, analysis, and development of methods for the classification or description of patterns, objects, signals, and processes. At the heart of this discipline is our ability infer the statistical behavior of data from limited data sets, and to assign data to classes based on generalized notions of distances in a probabilistic space.
8	Outline syllabus	CO Mapping
	Unit 1	**Introduction and mathematical preliminaries**
	A	Introduction to Pattern recognition; Applications areas in medical, defense, E-commerce, The Design Cycle. — CO1
	B	Clustering vs. Classification; Learning and Adaptation, Relevant basics of Linear Algebra. — CO1, CO2
	C	Vector spaces, Probability Theory, Estimation Theory. — CO1, CO2
	Unit 2	**Bayes Decision Theory**

	A	Data processing, Outliers, Correlation, Expectation, mean and covariance, classifiers	CO1, CO2, CO4
	B	Normal Distribution, Bayesian Classification, The Nearest- Neighbor Rule	CO2, CO4, CO5
	C	Introduction to Bayesian Decision Theory, Normal density and discriminant functions for the normal density	CO2, CO4
	Unit 3	**Clustering**	
	A	Basics of Clustering; similarity / dissimilarity measures; Criterion Functions for Clustering. Different distance functions and similarity measures, ,	CO1, CO2, CO3
	B	Clustering Techniques: K-means algorithm, Agglomerative hierarchical clustering	CO1, CO2, CO3
	C	K-medoids, DBSCAN, Cluster validation	CO1, CO2, CO3
	Unit 4	**Feature extraction and Feature selection**	
	A	Principal Component Analysis (PCA), Kernel PCA, Singular Value Decomposition, Fisher Linear discriminant analysis	CO2, CO3, CO5
	B	Algorithms - Branch and bound algorithm, sequential forward / backward selection algorithms, Maximum-Likelihood estimation,	CO2, CO3, CO5
	C	Probabilistic separability-based criterion functions, interclass distance-based criterion functions, K-Nearest Neighbor Estimation	CO2, CO3, CO5
	Unit 5	**Recent Advances in Patterns Recognitions**	
	A	Introduction to advanced pattern recognition schemes, Resources and tools used, Gaussian mixture models	CO2, CO3, CO6
	B	Support Vector Machine, Neural Networks, Hidden Markov Models (HMM),	CO2, CO3, CO6
	C	Basics to Biometrics: Biometric methodologies: finger prints, hand geometry, facial recognition, Iris scanning, retina scanning	CO2, CO3, CO6
	Mode of examination	Theory	
	Weightage Distribution	CA: 25% / MTE: 25% / ETE: 50%	
	Text book/s*	1. R. O. Duda, P. E. Hart and D. G. Stork, Pattern Classification, John Wiley, 2001. 2. Christopher M. Bishop, "Pattern Recognition and Machine Learning", Springer publication, 2006	
	Other References	3. S.Theodoridis and K.Koutroumbas, Pattern Recognition, 4th Ed., Academic Press, 2009. 4. Robert Schalkoff, "Pattern Recognition: Statistical, Structural and Neural Approaches", John Wiley & Sons, Inc.1992. 5. K.Jain, R.Bolle, S.Pankanti, "Biometric: Personal Identification in network society", Kluwer academic publishers, 1999.	

CO and PO Mapping

S. No.	Course Outcome	Program Outcomes (PO) & Program Specific Outcomes (PSO)
1.	Define Pattern concept and random process ideas, understand mathematical background.	
2.	Explain preliminary models to understand the solution environment.	
3.	Apply of existing mathematical solutions to test problems. and Perform Subspace analysis for classification problems	
4.	Classify patterns using Bayesian Decision Theory.	
5.	Evaluate patterns using Parametric and Non-Parametric techniques.	
6.	Discuss trajectory of recent trend in pattern recognition & understand various biometric technologies.	

PO and PSO mapping with level of strength for Course Name Pattern Recognition

S. No.	Course Outcome	Program Outcomes (PO) & Program Specific Outcomes (PSO)
1.	Define Pattern concept and random process ideas, understand mathematical background.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
2.	Explain preliminary models to understand the solution environment.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
3.	Apply of existing mathematical solutions to test problems. and Perform Subspace analysis for classification problems	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
4.	Classify patterns using Bayesian Decision Theory.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
5.	Evaluate patterns using Parametric and Non-Parametric techniques.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
6.	Discuss trajectory of recent trend in pattern recognition & understand various biometric technologies.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3

SHARDA SCHOOL OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE CODE	COURSE TITLE	L	T	P	C
CAL302	Pattern Recognition Lab	0	0	2	1

On successful completion of this module students will be able to:

CO 1. Define and Show naïve Bayesian Classifier for real world pro
CO 2. Classify patterns using Bayesian Decision Theory.
CO 3. Apply clustering techniques on read world problems
CO 4. Classify Feature extraction and Feature selection techniques.
CO 5. Evaluate patterns using Parametric and Non-Parametric techniques.
CO 6. Discuss trajectory of recent trend in pattern recognition & understand various biometric technologies.

LIST OF EXPERIMENTS

S No.	Experiment name	COURSE OUTCOME
1.	Introduction to Python, operations in Python	CO1
2.	Data Frames, Loading datasets Write a Python program to do the following operations: Data set Shared during lab (Example: xx_xx_xx.csv) Library: Pandas a) Loading data from CSV file b) Compute the basic statistics of given data - shape, no. of columns, mean c) Splitting a data frame on values of categorical variables d) Indexing of the data. e) Display top 10 rows of data	CO1
3.	Data preprocessing – handling missing values Write a python program to impute missing values with various techniques on given dataset. a) Remove rows/ attributes b) Replace with mean or mode c) Write a program to remove /replace missing values beyond a threshold value.	CO2
4.	Data Statistics and Data Visualization: Write a Python program to do the following operations: Data set Shared during lab (Example: xx_xx_xx.csv) Library: Pandas a) Display Frequency distribution b) Normalization c)Outlier detection d)Correlation analysis e) Different plots (bar plot, histogram, scatter plot etc)	CO2

5.	5.1 Classification: Write a program to implement the naïve Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets	**CO3**
	5.2 Clustering: Write a program to perform Clustering on a dataset using K-Means Algorithm. Display its Dendrogram.	**CO4**
6.	6.1 Write a program to implement PCA example using scikit-learn on Iris Data-set.	**CO3**
	6.2 Write a program to implement Nearest Neighbors classification on Iris Data-set and plot the decision boundaries for each class.	**CO4**
	6.3 Write a program to implement SVM for classification on standard dataset	**CO5**
	6.4 Write a program to implement Neural network for classification on standard dataset.	**CO6**
7.	**Value Added Program (Project)** Given a CSV file (Will be given in the lab), a) Perform Data pre-processing. b) Apply statistical analysis c) Perform Visual analytics d) On final data, perform Classification/ Clustering. e) Write a research paper and communicate the same in a journal or a conference.	**CO6**

Sign. of Course Coordinator	**Sign. of Lab. Coordinator**	**Sign. of HOD**
Ms. Kanika Singla	Prof. (Dr.) SHAJEE MOHAN	**Dr. Sudeep Varshney**
Name of Course Coordinator	**Name of Lab. Coordinator**	

School: SET		**Batch: 2023-27**	
Program: B-TECH		**Current Academic Year: 2023-24**	
Branch: CSE		**Semester:**	
1	**Course Code**	CSA303	
2	**Course Title**	**Deep Leaning and its Applications**	
3	**Credits**	**3**	
4	**Contact Hours (L-T-P)**	3-0-0	
	Course Status	CORE	
5	**Course Objective**	This course aims to present the mathematical, statistical and computational challenges of building stable representations for high-dimensional data, such as images, text and data. We will delve into selected topics of Deep Learning, discussing recent models from both supervised and unsupervised learning. Special emphasis will be on convolutional architectures, invariance learning, unsupervised learning and non-convex optimization.To understand and demonstrate how to solve general learning from a large series of data using computer based deep learning algorithms	
6	**Course Outcomes (CO's)**	On successful completion of this module students will be able to: 1. Recall Neural Networks relate it with Deep Learning concepts to solve real life applications 2. Compare and classify Regularization approaches for Deep Learning. 3. Build Convolutional Neural Networks models for image analysis and its applicability in societal problem solving. 4. Examine the Sequence models and analyse the relationships among them. 5. Assess the different Deep learning models based on their design processes. 6. Predict the behaviour of Deep learning models and apply them to solve real life applications.	
7	**Course Description**	This course starts with introduction to Deep Learning and further build, train, and deploy real world applications such as object recognition and Computer Vision, image and video processing, text analytics, Natural Language Processing, recommender systems, and other types of classifiers.	
8	**Syllabus Outline**		**CO Mapping**
	Unit 1	**Deep Feed forward Networks**	
	A	Recall Neural networks, Deep learning and its Practical aspects for real life applications ,Introduction to Simple Deep Neural Networks, Platform for Deep Learning, Deep Learning Software Libraries	CO1, CO6

	B	Introduction to Deep Feed Forward Networks ,Learning XOR, Gradient-Based Learning, Activation Functions, ReLU, Softmax, Sigmoid , Error Functions	CO1
	C	Architecture Design- Hidden Units Back-Propagation and Other Differentiation Algorithms	CO1
	Unit 2	**Regularization for Deep Learning**	
	A	Parameter Norm Penalties, Norm Penalties as Constrained Optimization, Regularization and Under-Constrained Problems, Dataset Augmentation, Noise Robustness, Semi-Supervised Learning, Multitask Learning, Early Stopping, Parameter Tying and Parameter Sharing, Bagging , Drop Out, Difficulty of training deep neural networks, Greedy layer wise training, Adversarial Training	CO2
	B	How Learning Differs from Pure Optimization, Challenges in Neural Network Optimization, Basic Algorithms: Stochastic Gradient Descent, Momentum, Nesterov Momentum Parameter Initialization Strategies Algorithms with Adaptive Learning Rates, AdaGrad. RMSProp. Adam	CO2
	C	Introduction to Autoencoder, Undercomplete Autoencoder, Regularized Autoencoders, Representational Power, Layer Size and Depth. Stochastic Encoders and Decoders, Applications of Encoder Decoder models	CO2
	Unit 3	**Convolutional Neural Networks**	
	A	Why CNN?, Its role, significance and applicability in societal problem solving , The Convolution Operation, Motivation, Pooling, The Neuroscientific Basis for Convolutional Networks	CO1, CO3, CO6
	B	Prior probability distribution, Convolution and Pooling as an Infinitely Strong Prior, Variants of the Basic Convolution Function, Structured Outputs, Data types with different dimensionalities and number of channel	CO1, CO3
	C	Efficient Convolution Algorithms, Random or Unsupervised Features of CNN , Normalization, Applications of CNN in Computer Vision – ImageNet	CO1, CO3, CO6
	Unit 4	**Sequence Modeling: Recurrent Neural Networks**	
	A	Sequence Learning Problems , Recurrent Neural Network and its significance in real world, RNN model, Backpropagation through time ,Bidirectional RNNs	CO4, CO6
	B	Different types of RNNs, Gated Recurrent Unit (GRU) Recursive Neural Networks , The Challenge of Long-Term Dependencies	CO4
	C	Introduction of Long Short Term Memory Neural Networks, Learning Algorithm of LSTM/ RNN Bidirectional LSTMs	CO4
	Unit 5	**Deep Networks and design process**	

	A	Introduction to Generative Adversarial Networks , Generative Adversarial Networks – Architecture,			CO5,CO6
	B	DCGAN, GAN hack, Applications of Generative Adversarial Networks			CO5,CO6
	C	Practical design process for deep learning techniques based on real world problems: Performance Metrics , Default Baseline Models, Determining Whether to Gather More Data, Selecting Hyperparameters, Debugging Strategies			CO5,CO6
	Mode of examination	Theory			
	Weightage Distribution	**CA**	**MTE**	**ETE**	
		25%	25%	50%	
	Text Books	1. **Deep Learning, by Goodfellow I., Bengio Y. & Courville A. (2016)** 2. **Visualizing and Understanding Convolutional Networks, by Matt Zeiler, Rob Fergus** 3. **TensorFlow: a system for large-scale machine learning, by Martín A., Paul B., Jianmin C., Zhifeng C., Andy D. et al. (2019)**			
	Reference Books	1. Deep learning in neural networks, **by JuergenSchmidhuber (2015)** 2. https://cs230.stanford.edu/syllabus/ 3. https://towardsdatascience.com/september-edition-machine-learning-case-studies-a3a61dc94f23 4. **Deep Learning: A Practitioner's Approach by Josh Patterson, Oreilly.**			

CO and PO Mapping

S. No.	Course Outcome (CO)	Program Outcomes (PO) & Program Specific Outcomes (PSO)
1	Recall Neural Networks relate it with Deep Learning concepts to solve real life applications	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
2	Compare and classify Regularization approaches for Deep Learning.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
3	Build Convolutional Neural Networks models for image analysis and its applicability in societal problem solving.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
4	Examine the Sequence models and analyse the relationships among them.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
5	Assess the different Deep learning models based on their design processes.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
6	Predict the behaviour of Deep learning models and apply them to solve real life applications.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3

PO and PSO mapping with level of strength for Course Name: Introduction to Deep Leaning

(Course Code- CSA302)

School: SET		**Batch : 2023-27**
Program: B-TECH		**Current Academic Year: 2023-24**
Branch: CSE		**Semester:**
1	**Course Code**	CAL303
2	**Course Title**	**Deep Leaning and its Applications Lab**
3	**Credits**	**1**
4	**Contact Hours (L-T-P)**	0-0-2
	Course Status	CORE
5	**Course Objective**	This course aims to present the mathematical, statistical and computational challenges of building stable representations for high-dimensional data, such as images, text and data. We will delve into selected topics of Deep Learning, discussing recent models from both supervised and unsupervised learning. Special emphasis will be on convolutional architectures, invariance learning, unsupervised learning and non-convex optimization. To understand and demonstrate how to solve general learning from a large series of data using computer based deep learning algorithms
6	**Course Outcomes (CO's)**	On successful completion of this module students will be able to: 1. Define and show the implementation of Deep Learning concepts to solve real life applications 2. Compare and classify Regularization approaches for Deep Learning. 3. Build Convolutional Neural Networks models for image analysis and its applicability in societal problem solving. 4. Examine the Sequence models and analyses the relationships among them. 5. Assess the different Deep learning models based on their design processes. 6. Predict the behavior of Deep learning models and apply them to solve real life applications.
7	**Course Description**	This course starts with introduction to Deep Learning and further build, train, and deploy real world applications such as object recognition and Computer Vision, image and video processing, text analytics, Natural Language Processing, recommender systems, and other types of classifiers.

8	Outline syllabus		CO Mapping
	Unit 1	**Deep Feed Forward Networks**	
	1	Write a program to implement Deep Feed Forward Network to predict Housing price available on Kaggle.	**CO1**
	2	Write a program to implement classification using Deep Feed Forward Network on dataset available on Kaggle.	**CO1**
	Unit 2	**Regularization for Deep Learning**	

	3	Write a program to implement regularization to overcome over fitting problem in Housing price Prediction	CO2
	Unit 3	**Convolutional Neural Networks**	
	4	**Digit Recognition from** MNIST : Given a zip codes hand written on envelops, identify the digit for each hand written character. A model of this problem would allow a computer program to read and understand handwritten zip codes and sort envelops by geographic region.	CO1, CO3, , CO6
	5	Dog-Breed Classifier: Design and train a convolutional neural network to analyze images of dogs and correctly identify their breeds. Use transfer learning and well-known architectures to improve this model—this is excellent preparation for more advanced applications.	CO1, CO3
	Unit 4	**Sequence Modelling: Recurrent Neural Networks**	
	6	**Stock market prediction on** NASDAQ stocks: Given the current and past price movements for a stock, determine whether the stock should be bought, held or sold. A model of this decision problem could provide decision support to financial analysts.	CO4, CO6
	7	The Slot-Filling (Spoken Language Understanding) consists in assigning a label to each word given a sentence. It's a classification task.	CO4
	8	Write a program to implement credit card fraud detection prediction	CO4
	Unit 5	**Deep Networks and design process**	
	9	Implement Convolutional Autoencoders in Python with Keras on MNIST dataset	CO5,CO6
	10	Write a program to implement Generative Adversarial Network	CO5,CO6
	Mode of examination	Practical	
	Weightage Distribution	CA: 25% / MTE: 25% / ETE: 50%	
	Text book/s*	1. Lipschutz, "Data Structures" Schaum's Outline Series, TMH	
	Other References	1. Aaron M. Tenenbaum, Yedidyah Langsam and Moshe J. Augenstein "Data Structures Using C and C++" , PHI 2. Horowitz and Sahani, "Fundamentals of Data Structures", Galgotia Publication 3. Jean Paul Trembley and Paul G. Sorenson, "An Introduction to Data Structures with applications", McGraw Hill 4. R. Kruse etal, "Data Structures and Program Design in C", Pearson Education 5. G A V Pai, "Data Structures and Algorithms", TMH	
	Weblink	**https://towardsdatascience.com/getting-rich-quick-with-machine-learning-and-stock-market-predictions-696802da94fe**	

		https://www.datacamp.com/community/tutorials/autoencoder-keras-tutorial http://deeplearning.net/tutorial/rnnslu.html	

CO and PO Mapping

S. No.	Course Outcome (CO)	Program Outcomes (PO) & Program Specific Outcomes (PSO)
1	Define and show the implementation of Deep Learning concepts to solve real life applications	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
2	Compare and classify Regularization approaches for Deep Learning.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
3	Build Convolutional Neural Networks models for image analysis and its applicability in societal problem solving.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
4	Examine the Sequence models and analyse the relationships among them.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
5	Assess the different Deep learning models based on their design processes.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
6	Predict the behaviour of Deep learning models and apply them to solve real life applications.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3

PO and PSO mapping with level of strength for Course Name: Introduction to Deep Leaning

(Course Code- CAL303)

Course Code_ Course Name	CO's	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
Deep Leaning and its Applications Lab (Course Code- CAL303)	CO1	3	3	3	3	2	1	1	1	1	3	1	3	3	3	1
	CO2	3	3	3	3	3	3	3	1	3	3	3	3	3	3	3
	CO3	3	3	3	3	3	2	2	1	2	3	3	3	3	3	3
	CO4	3	3	3	3	3	2	2	1	2	3	3	3	3	3	3
	CO5	3	3	3	3	3	3	3	1	3	3	3	3	3	3	3
	CO6	3	3	3	3	3	3	3	2	3	3	3	3	3	3	3

Average of non-zeros entry in following table (should be auto calculated).

Course Code	Course Name	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CAL 303	**Deep Leaning and its Applications Lab**	3.00	3.00	3.00	3.00	2.83	2.33	2.33	1.17	2.33	3.00	2.67	3.00	3.00	3.00	2.67

Total 40.3

Strength of Correlation

1. Addressed to **Slight (Low=1) extent** **2.** Addressed to **Moderate (Medium=2) extent**
3. Addressed to **Substantial (High=3) extent**

School: SET		**Batch: 2023-27**	
Program: B.Tech		**Current Academic Year: 2023-24**	
Branch: ALL		**Semester: VII**	
1	**Course Code**	**CSA401**	**Course Name: Computer Vision**
2	**Course Title**	**Computer Vision**	
3	**Credits**	**3**	
4	**Contact Hours (L-T-P)**	**3-0-0**	
	Course Status	**Program Elective**	
5	Course Objective	1. To implement fundamental image processing techniques required for computer vision 2. To develop applications using computer vision techniques	
6	Course Outcomes	Students will be able to have thorough Understanding of: CO-1 Define the Fundamentals of Computer Vision and Computer Graphics and relate them with real world applications CO-2 Explain Image formation models and Foundations for Mathematical basis for various Projection Systems CO- 3 Apply Image processing techniques such as Segmentation and Edge Detection for real time and real world applications. CO- 4 Analyze various feature extraction techniques for different problem domain. CO-5 Evaluate Pattern Recognition Using Clustering, Classification, Supervised Learning and Unsupervised Learning Techniques CO-6 Build computer vision applications for real world Applications.	
7	Course Description	In this course students will learn basic principles of image formation, image processing algorithms, extracting the features and then analyzing the underlying patterns.	
8	Outline syllabus		CO Mapping
	Unit 1	**Introduction to Computer Vision**	
	A	Computer Vision and Computer Graphics , What is Computer Vision - Low-level, Mid-level, High-level	CO1
	B	Overview of Diverse Computer Vision Applications: Document Image Analysis, Biometrics, Object Recognition, Tracking, Medical Image Analysis	CO1
	C	Face detection, Face recognition, Eigen faces, Active appearance and 3D shape models of faces, Surveillance, foreground-background separation, vehicle vision system: locating roadway, road markings, identifying road signs, locating pedestrians	CO1
	Unit 2	**Image Formation Models**	
	A	Monocular imaging system , Radiosity: The 'Physics' of Image Formation, Radiance, Irradiance, Brightness, color etc,	CO2

	B	Orthographic & Perspective Projection ,Camera model and Camera calibration, Binocular imaging systems			CO2
	C	Multiple views geometry, Structure determination, shape from shading, Weak perspective projection and orthographic projection, Concept of image coordinate system and camera coordinate system;			CO2
	Unit 3	**Image Processing**			
	A	Image preprocessing: The Discrete Fourier Transform (DFT) of Two Variables, Properties of the 2-D DFT, Discrete Cosine Transform (DCT)			CO3, CO6
	B	Wavelet Transforms in One Dimension-The Discrete Wavelet Transform (DWT) and The Continuous Wavelet Transform. Wavelet Decomposition,			CO3, CO6
	C	Orthogonal, Euclidean, Affine, Projective, etc; Convolution and Filtering, Image Enhancement, Restoration, Histogram Processing.			CO3, CO6
	Unit 4	**Image Processing Operations**			
	A	Image Filtering (spatial domain), Mask-based (e.g., correlation, convolution), Smoothing (e.g., Gaussian), Sharpening (e.g., gradient)			CO4
	B	Segmentation : Edge-based (e.g., voting, optimization, perceptual grouping), Pixel-based (e.g., clustering)			CO4
	C	Colour fundamentals, Colour models, Colour transformation, Smoothing and Sharpening, Colour segmentation			CO4
	Unit 5	**Feature Extraction**			
	A	Edge detection: Canny, Laplacian of Gaussian; Line detectors (Hough Transform)			CO5, CO6
	B	Corners - Harris and Hessian Affine, Orientation Histogram, SIFT, SURF, HOG, GLOH			CO5, CO6
	C	Scale-Space Analysis- Image Pyramids and Gaussian derivative filters, Gabor Filters			CO5, CO6
	Mode of examination	Theory			
	Weightage Distribution	CA	MTE	ETE	
		25%	25%	50%	
	Text book/s*	1. Milan Sonka, Vaclav Hlavac, Roger Boyle," Digital Image Processing and Computer Vision" Cengage Learning, 1st Edition, 2008 2. Computer Vision - A modern approach, by D. Forsyth and J. Ponce, Prentice Hall Robot Vision, by B. K. P. Horn, McGraw-Hill.			
	Reference Books	1, Introductory Techniques for 3D Computer Vision, by E. Trucco and A. Verri, Publisher: Prentice Hall. 2. R. C. Gonzalez, R. E. Woods. Digital Image Processing. Addison Wesley Longman, Inc., 1992. 3. D. H. Ballard, C. M. Brown. Computer Vision. Prentice-Hall, Englewood Cliffs, 1982.			

CO and PO Mapping

S. No.	Course Outcome	Program Outcomes (PO) & Program Specific Outcomes (PSO)
1.	CO-1 Define the Fundamentals of Computer Vision and Computer Graphics and relate them with real world applications	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
2.	CO-2 Explain Image formation models and Foundations for Mathematical basis for various Projection Systems	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
3.	CO- 3 Apply Image processing techniques such as Segmentation and Edge Detection for real time and real world applications.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
4.	CO- 4 Analyze various feature extraction techniques for different problem domain.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
5.	CO-5 Evaluate Pattern Recognition Using Clustering, Classification, Supervised Learning and Unsupervised Learning Techniques	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
6.	CO-6 Build computer vision applications for real world applications.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3

PO and PSO mapping with level of strength for Course Name Computer Vision **(Course Code** CSA-401**)**

Subject	PO's / PSO's	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
Computer Vision CSA-401	CO1	3	3	3	3	1	1	1	1	1	2	1	3	2	3	1
	CO2	3	3	3	3	2	1	1	1	1	2	1	3	2	3	2
	CO3	3	3	3	3	2	1	1	1	1	2	1	3	3	3	2
	CO4	3	3	3	3	2	2	1	1	1	2	1	3	3	3	2
	CO5	3	3	3	3	2	2	1	1	1	2	1	3	3	3	2
	CO6	3	3	3	3	2	3	3	1	3	2	1	3	3	3	3

Average of non-zeros entry in following table (should be auto calculated).

Course Code	Course Name	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CSA-401	Computer Vision	3.00	3.00	3.00	3.00	1.83	1.67	1.33	1.00	1.33	2.00	1.00	3.00	2.67	3.00	2.00

Total- 32.83
Strength of Correlation

1. Addressed to **Slight (Low=1) extent** **2.** Addressed to **Moderate (Medium=2) extent**
3. Addressed to **Substantial (High=3) extent**

School: SET		**Batch: 2023-27**	
Program: B.Tech		**Current Academic Year: 2023-24**	
Branch: ALL		**Semester: VII**	
1	**Course Code**	**CAL401**	**Course Name: Computer Vision Lab**
2	**Course Title**	**Computer Vision Lab**	
3	**Credits**	**1**	
4	**Contact Hours (L-T-P)**	**0-0-2**	
	Course Status	**Core**	
5	Course Objective	To implement fundamental image processing techniques required for computer vision To develop applications using computer vision techniques	
6	Course Outcomes	Students will be able to have thorough Understanding of: CO-1 Define and show the Fundamentals of Computer Vision techniques on images CO-2 Show the Image filtering and opening / closing operations on Color images CO- 3 Apply Image transformation techniques such as for real time and real world applications. CO- 4 Analyze various feature extraction techniques for different Problem domains. CO-5 Evaluate Pattern Recognition Using Clustering, Classification Techniques CO-6 Build computer vision applications for real world Problems.	
7	Course Description	In this course students will learn basic principles of image formation, image processing algorithms, extracting the features and then analyzing the underlying patterns.	
8	Outline syllabus		CO Mapping
	Unit 1	**Introduction to Computer Vision**	
	1	To create a program to display grayscale image using read and write operation.	CO1
	2	To create a vision program to find histogram value and display histograph of a grayscale and color image.	CO1
		Write a program for color image processing	
	Unit 2	**Image Formation Models**	
	3	To Implement smoothing or averaging filter in spatial domain	CO2
	4	Program for opening and closing of the image.	CO2
	5	To fill the region of interest for the image	CO2
	Unit 3	**Image Processing**	
	6	To create a vision program for Non-Linear Filtering technique using edge detection	CO3, CO6
	7	To create a program to discretize an image using Fourier transformation.	CO3, CO6
	8	To create a vision program to determine the edge detection of an image using different operators.	CO3, CO6
	Unit 4	**Feature Extraction**	
	9	Program of sharpen image using gradient mask.	CO4

	10	Program for morphological operation: erosion and dilation.			CO4
	11	Write a program for image segmentation using local and global thresholding			CO4
	Unit 5	**Pattern Analysis**			
	12	Write a program to implement image classification.			CO5, CO6
	13	Write a program to implement image clustering.			CO5, CO6
	Mode of examination	Lab			
	Weightage Distribution	CA	MTE	ETE	
		25%	25%	50%	
	Text book/s*	1. Milan Sonka, Vaclav Hlavac, Roger Boyle," Digital Image Processing and Computer Vision" Cengage Learning, 1st Edition, 2008 2. Computer Vision - A modern approach, by D. Forsyth and J. Ponce, Prentice Hall Robot Vision, by B. K. P. Horn, McGraw-Hill.			
	Reference Books	1, Introductory Techniques for 3D Computer Vision, by E. Trucco and A. Verri, Publisher: Prentice Hall. 2. R. C. Gonzalez, R. E. Woods. Digital Image Processing. Addison Wesley Longman, Inc., 1992. 3. D. H. Ballard, C. M. Brown. Computer Vision. Prentice-Hall, Englewood Cliffs, 1982.			

CO and PO Mapping

S. No.	Course Outcome	Program Outcomes (PO) & Program Specific Outcomes (PSO)
1.	CO-1 Define and show the Fundamentals of Computer Vision techniques on images	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
2.	CO-2 Show the Image filtering and opening / closing operations on Color images	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
3.	CO- 3 Apply Image transformation techniques such as for real time and real world applications.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
4.	CO- 4 Analyze various feature extraction techniques for different Problem domains.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
5.	CO-5 Evaluate Pattern Recognition Using Clustering, Classification Techniques	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
6.	CO-6 Build computer vision applications for real world Problems.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3

PO and PSO mapping with level of strength for Course Name Computer Vision **(Course Code** CSA-301**)**

School SET		**Batch: 2023-27**	
Program B-TECH		**Current Academic Year: 2023-24**	
Branch CSE		**Semester VI**	
1	Course Code	**CSA041**	Course Name **Natural Language Processing**
2	Course Title	**Natural Language Processing**	
3	Credits	3	
4	Contact Hours (L-T-P)	**3-0-0**	
	Course Status	**Specialization Elective**	
5	Course Objective	**Students will try to learn** 3. Basics of natural language processing. 4. How to apply basic algorithms in natural language processing. 5. Algorithmic description of the main language levels morphology, syntax, semantics, and pragmatics. 6. Basics of knowledge representation, inference, and relations to the artificial intelligence. 7. Techniques such as tokenization, stemming, and lemmatization.	
6	Course Outcomes	Students will be able to CO-1. **Define** Computational Linguistics phenomena and making decisions using it. CO-2. **Explain** how to access Text Corpora and Lexical Resources. CO-3. **Apply** processing of raw text using NLP programming concepts. CO-4. **Analyze** tagging of words and Extracting Information from Text. CO-5. **Discuss** analysis of sentences using CFG and Propositional Logic. CO-6. **Design** NLP based applications for different business environment.	
7	Course Description	This course provides an introduction to the field of computational linguistics, aka natural language processing (NLP). We will learn how to create systems that can understand and produce language, for applications such as information extraction, machine translation, automatic summarization, question-answering, and interactive dialogue systems. The course will cover linguistic (knowledge-based) and statistical approaches to language processing in the three major subfields of NLP: syntax (language structures), semantics (language meaning), and pragmatics/discourse (the interpretation of language in context).	
8	Outline syllabus		CO Mapping
	Unit 1	**Introduction and Computational Linguistics**	
	A	What is Natural Language Processing, hands-on demonstrations. Ambiguity and uncertainty in language. The Turing test	CO1
	B	Computing with Language Texts and Words Implementation of NLTK, Searching Text, Counting Vocabulary, A Closer Look at Python Texts as Lists of Words, Computing with Language Simple Statistics, Frequency Distributions, Fine-grained Selection of Words, Collocations and Bigrams,	CO1

	C	Making Decisions and Taking Control, Conditionals, Operating on Every Element, Nested Code Blocks, Looping with Conditions	CO1
	Unit 2	**Accessing Text Corpora and Lexical Resources**	
	A	Automatic Natural Language Understanding, Word Sense Disambiguation, Pronoun Resolution, Generating Language Output, Machine Translation, Spoken Dialog Systems, Textual Entailment, Limitations of NLP	CO2
	B	Accessing Text Corpora, Gutenberg Corpus, Web and Chat Text, Brown Corpus, Reuters Corpus, Inaugural Address Corpus, Annotated Text Corpora, Corpora in Other Languages	CO2
	C	Text Corpus Structure, Loading your own Corpus, Conditional Frequency Distributions, Conditions and Events, Counting Words by Genre, Plotting and Tabulating Distributions, Generating Random Text with Bigrams,	CO2
	Unit 3	**Processing Raw Text**	
	A	Lexical Resources,, Wordlist Corpora, A Pronouncing Dictionary, Comparative Wordlists, Shoebox and Toolbox Lexicons, WordNet, Senses and Synonyms, The WordNet Hierarchy, Lexical Relations, Semantic Similarity	CO3
	B	Accessing Text from the Web and from Disk, Strings Text Processing at the Lowest Level, Text Processing with Unicode, Regular Expressions for Detecting Word Patterns, Useful Applications of Regular Expressions Normalizing Text, Regular Expressions for Tokenizing Text, Segmentation,Formatting From Lists to Strings	CO3
	C	NLP Programming Sequences, Style, Functions for text processing, Program Development & Algorithm Design using, Python Libraries	CO3
	Unit 4	**Tagging & Information Extraction**	
	A	Categorizing and Tagging Words Using a Tagger,Tagged Corpora,Mapping Words to Properties Using Python Dictionaries, Automatic Tagging, N-Gram Tagging,Transformation-Based Tagging, Determine the Category of a Word	CO4
	B	Text classification Supervised Classification, Examples of Supervised Classification,Evaluation,Decision Trees,Naive Bayes Classifiers,Maximum Entropy Classifiers,Modeling Linguistic Patterns	CO4
	C	Extracting Information from Text Information Extraction,Chunking,Developing and Evaluating Chunkers,Recursion in Linguistic Structure,Named Entity Recognition,Relation Extraction	CO4
	Unit 5	**Analysis of sentences**	
	A	Analyzing Sentence Structure Grammatical Dilemmas,What's the Use of Syntax?,Context-Free Grammar,Parsing with Context-Free Grammar,Dependencies and Dependency Grammar,Grammar Development	CO5

	B	Analyzing the Meaning of Sentences Natural Language Understanding,Propositional Logic,First-Order Logic,The Semantics of English Sentences,Discourse Semantics			CO5
	C	Managing Linguistic DataCorpus Structure,The Life Cycle of a Corpus,Acquiring Data,Working with XML,Working with Toolbox Data,Describing Language Resources Using OLAC Metadata			CO5
	Mode of examination	Theory			
	Weightage Distribution	CA	MTE	ETE	
		25%	25%	50%	
	Text book/s*	1. Speech and Language processing an introduction to Natural Language Processing, Computational Linguistics and speech Recognition by Daniel Jurafsky and James H. Martin (ISBN13: 978-0131873216) 2. Ruslan Mitkov, The Oxford Handbook of Computational Linguistics, Oxford University Press, 2005			
	Other References	3. Charu C. Aggarwal and Cheng Xiang Zhai, Mining Text Data, Springer, 2012 4. Hopcroft, J.E. and Ullman, J.D., Introduction to Automata, Theory and Languages, Addison-Wesley, 1979			

CO and PO Mapping

S. No.	Course Outcome	Program Outcomes (PO) & Program Specific Outcomes (PSO)
1.	**Define** Computational Linguistics phenomena and making decisions using it.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
2.	**Explain** how to access Text Corpora and Lexical Resources.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
3.	**Apply** processing of raw text using NLP programming concepts.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
4.	**Analyze** tagging of words and Extracting Information from Text.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
5.	**Discuss** analysis of sentences using CFG and Propositional Logic.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
6.	**Design** NLP based applications for different business environment.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3

School: SET		**Batch: 2023-27**	
Program: B.Tech.		**Current Academic Year: 2023-24**	
Branch: CSE		**Semester: VII**	
1	**Course Code**	**CSA051**	**Course Name- RECOMENDER SYSTEMS**
2	**Course Title**	**RECOMENDER SYSTEMS**	
3	**Credits**	**3**	
4	**Contact Hours (L-T-P)**	**3-0-0**	
	Course Status	**Specialization Elective**	
5	Course Objective	To develop state-of-the-art recommender systems that automate a variety of choice-making strategies with the goal of providing affordable, personal, and high-quality recommendations 1. To introduce fundamental techniques in recommender systems. 2. To introduce the ideas of Non-personalized and project-association recommenders through content-based and collaborative techniques. 3. To become familiar with to various approaches for building recommender systems including collaborative, content-based, knowledge-based, and hybrid methods.	
6	Course Outcomes	After Successful completion of this course the student will be able to: CO-1. **Define** the basics of Recommender Systems and its types**.** CO-2. **Explain** the similarity measures used in formation of neighbourhood of samples of data. CO-3. **Apply** various techniques of content and knowledge based recommendation for real life applications. CO-4. **Analyse** and categorize the various recommendation techniques for hybridization. CO-5. **Choose** the suitable type of Recommender systems for societal problems CO-6. **Design** the recommender system to support all online applications of folksonomies and Social Networking sites.	
7	Course Description	Recommender systems offer personalized access to online information in product catalogs, social media networks, and document collections, among other applications. It will introduce students to various approaches for building recommender systems including collaborative, content-based, knowledge-based, and hybrid methods.	
8	Outline syllabus		CO Mapping
	Unit 1	**Introduction**	
	A	Introduction to Recommender Systems, Neighbourhood-based methods Recommendation, Applications of recommender systems, Case study of movie lens, group lens and amazon.com etc.	CO1,

	B	Introduction to Information retrieval, Introduction to collaborative filtering			CO1,
	C	Knowledge sources, Neighbourhood-based methods.			CO1
	Unit 2	**Memory and Model-based Collaborative Recommendation**			
	A	Similarity measures used in Collaborative Filtering, Model-based Collaborative Recommendation Dimensionality reduction. Regression: Slope1 and SLIM models. Association rules and Naïve Bayes models,			CO2, CO6
	B	Factorization Methods of Collaborative Recommendation, Latent factor models.			CO2, CO6
	C	Optimization techniques. Singular value decomposition, constrained matrix factorization.			CO2, CO6
	Unit 3	**Content-based and Knowledge-based Recommendation**			
	A	High level architecture of content-based systems, Advantages and drawbacks of content based filtering, Content-based Recommendation Feature representation, extraction, and selection.			CO3, CO6
	B	User profiles. Learning models, Item profiles, Discovering features of documents, Obtaining item features from tags			CO3, CO6
	C	Knowledge-based Recommendation Constraint-based recommendation. Critiquing systems.			CO3, CO6
	Unit 4	**Hybrid recommendation and Evaluation**			CO3, CO6
	A	Hybrid Recommendation Complementarities between recommendation techniques and knowledge sources.			CO3, CO6
	B	Combining recommendation methods. Types of evaluation for recommender systems			CO3, CO6
	C	Evaluation design. Prediction metrics and ranking metrics. A/B Testing			CO3, CO6
	Unit 5	**Context-aware recommendation**			
	A	Context effects in recommendation. Types and representations of context.			CO5, CO6
	B	Pre-filtering, post-filtering and contextual modelling, Temporal and location-sensitive models			CO5, CO6
	C	Recommender Systems and communities: Communities, collaboration and recommender systems in personalized web search, Social tagging recommender systems, Trust and Recommendations, Group recommender systems and their applications in solving societal problems.			CO5, CO6
	Mode of examination	Theory			
		CA	MTE	ETE	

	Weightage Distribution	25%	25%	50%	
	Text book/s*	Aggarwal, C. C. Recommender Systems: The Textbook. Springer 2019. ISBN 978-3-319-29657-9. Available through the DePaul library.			
	Other References	http://www.deitel.com/ResourceCenters/Web20/RecommenderSystems/RecommenderSystemsCourseSyllabi/tabid/1321/Default.aspx			
	Other References	Francesco Ricci, LiorRokach and BrachaShapira Recommender Systems Handbook, 2005			

CO and PO Mapping

S. No.	Course Outcome	Program Outcomes (PO) & Program Specific Outcomes (PSO)
1.	**Define** the basics of Recommender Systems and its types**.**	PO1,PO2,PO3,PO4, PO5,PO6, PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
2.	**Explain** the similarity measures used in formation of neighbourhood of samples of data.	PO1,PO2, PO3,PO4,PO5,PO6, PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
3.	**Apply** various techniques of content and knowledge based recommendation for real life applications.	PO1,PO2,PO3,PO4,PO5,PO6, PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
4.	**Analyse** and categorize the various recommendation techniques for hybridization.	PO1,PO2,PO3,PO4,PO5,PO6, PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
5.	**Choose** the suitable type of Recommender systems for societal problems	PO1,PO2,PO3,PO4, PO5,PO6, PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
6.	**Design** the recommender system to support all online applications of folksonomies and Social Net. sites.	PO1,PO2,PO3,PO4, PO5,PO6, PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3

PO and PSO mapping with level of strength for RECOMENDER SYSTEMS (Course Code CSA051)

Subject	Course Objectives	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
RECOMENDER SYSTEMS CSA-051	CO1	3	3	2	3	2	1	1	1	2	1	-	3	3	2	2
	CO2	3	3	3	3	3	2	2	1	2	2	-	3	3	3	2
	CO3	3	3	3	3	3	3	3	1	3	2	-	3	3	2	2
	CO4	3	3	3	3	3	2	2	1	3	2	-	3	3	3	2
	CO5	3	3	3	3	3	3	3	1	3	2	-	3	3	3	2
	CO6	3	3	3	3	3	3	3	1	3	3	-	3	3	3	3

Average of non-zeros entry in following table (should be auto calculated).

Course Code	Course Name	PO 1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CSA-051	**RECOMENDER SYSTEMS**	**3.00**	**3.00**	**2.83**	**3.00**	**2.83**	**2.33**	**2.33**	**1.00**	**2.67**	**2.00**	**0.00**	**3.00**	**3.00**	**2.67**	**2.17**

Total- 35.83

School: SET		**Batch: 2023-27**	
Program: B.Tech		**Current Academic Year: 2023-24**	
Branch: CSE / IT		**Semester: 6th**	
1	Course Code	**CSP398**	Course Name: Project Based Learning -4
2	Course Title	Project Based Learning -4	
3	Credits	2	
4	Contact Hours (L-T-P)	0-0-4	
	Course Status	Compulsory	
5	Course Objective	1. To align student's skill and interests with a realistic problem or project. 2.To understand the significance of problem and its scope. 3.Students will make decisions within a framework.	
6	Course Outcomes	Students will be able to: CO1: Identify and formulate problem statement. CO2: Design relational database schema. CO3: Develop the solution by using different aspects of programming language. CO4: Classify and understand various test techniques for verification and validation of project. CO5: Analyze and make use of modern for solving real word problems. CO6: Develop teamwork and need to engage in life-long learning, along with the ability to communicate effectively with others.	
7	Course Description	In PBL-4, the students will learn how to define the problem for developing projects, and Design applicable solutions in one or more application domains using software engineering approaches that integrate ethical, social, legal and economic concerns.	
8	Outline syllabus		CO Mapping
	Unit 1	Problem Definition and identification, Team/Group formation and Project Assignment. Finalizing the problem statement, resource requirement, if any.	CO1,CO4
	Unit 2	Use of the relational algebra operations from mathematical set theory (union, intersection, difference, and Cartesian product) and the relational algebra operations developed specifically for relational databases (select (restrict), project, join, and division)..	CO2,CO6
	Unit 3	Design; implement project work in any programming language.	CO3
	Unit 4	Use of various test tools and techniques for software verification and validation of project	CO4,CO5
	Unit 5	Demonstrate and execute Project with the team.	CO6
		Report should include Abstract, Hardware / Software Requirement, Problem Statement, Design/Algorithm, ER diagrams, Use Case Diagrams, State Diagrams, Sequence Diagrams, Communication Diagrams, and	

		Activity Diagrams, Implementation Detail. Validation Reports. References, Test cases if any. The presentation, report, work done during the term supported by the documentation, forms the basis of assessment.			
	Mode of examination	Practical /Viva			
	Weight age Distribution	CA			MTE
			CE(Viva)	ETE	
		25%	25%	50%	

CO and PO Mapping

S. No.	Course Outcome	Program Outcomes (PO)
1.	CO1: Identify and formulate problem statement.	PO1, PO2, PO4,PO6, PO8,PO9, PO10, PO11, PO12,PSO1,PSO2,PSO3
2.	CO2: Design relational database schema.	PO1, PO2, PO3,PO4,PO5, PO7, PO8, PO9, PO11, PO12 , PSO1,PSO2,PSO3
3.	CO3: Develop the solution by using different aspects of programming language.	PO1, PO2, PO3,PO4,PO5, PO6, PO8, PO9, PO11, PO12, PSO1,PSO2
4.	CO4: Classify and understand various test techniques for verification and validation of project.	PO1, PO2, PO3,PO4,PO5, PO8,PO9, PO10, PO11, PO12 ,PSO1,PSO2,PSO3
5.	CO5: Analyze and make use of modern for solving real word problems.	PO1, PO2, PO5, PO6, PO7, PO8, PO9, PO12 PSO1,PSO2
6.	CO6: Develop teamwork and need to engage in life-long learning, along with the ability to communicate effectively with others.	PO2, PO4, PO8,PO9, PO10, PO11, PO12,PSO1,PSO3

School: SSCSE		**Batch: 2025-2029**	
Program: B.Tech		**Current Academic Year: 2025-26**	
Branch: CSE(AR/VR)		**Semester: VI**	
1	**Course Code**	**CSHC3102**	**Course Name: Brain Computer Interface**
2	**Course Title**	**Brain Computer Interface**	
3	**Credits**	**2**	
4	**Contact Hours (L-T-P)**	**2-0-0**	
	Course Status	**Specialization Elective**	
5	Course Objective	1. Develop the ability to design, 2. Implement and manipulate Brain Computer Interface technology. 3. Introduce concepts of Data Streaming and Data Processing using suitable tools. 4. Apply Brain Computer Interface technology concepts to various examples and real-life applications.	
6	Course Outcomes	After the completion of this course students will be able to: **CO1:** Understand the fundamental concepts of Brain-Computer Interface (BCI), its applications, and historical development. **CO2:** Demonstrate the different methods of acquiring brain signals such as EEG, ECoG, and MEG. **CO3:** Implement classification techniques using machine learning algorithms for BCI signal analysis. **CO4:** Analyze ethical, legal, and societal issues related to the development and deployment of BCI technologies.. **CO5:** Assess the Recent Trends and Future Directions in BCI. **CO6:** Design an experiment to test a perceptual phenomenon.	
7	Course Description	To provide students with a solid understanding of the basic concepts, principles, and technologies behind BCIs, including brain signal acquisition and processing techniques. This course provides hands-on experience in designing and implementing Brain-Computer Interface systems, such as controlling prosthetic devices, communication aids, and assistive technologies. Students will gain latest research, advancements, and future directions in the field of Brain-Computer Interfaces, including the integration of AI and potential applications in healthcare, entertainment, and human augmentation.	
8	Outline syllabus		CO Mapping
	Unit 1	**Introduction to Brain-Computer Interface (BCI)**	
	A	Overview of Brain-Computer Interface (BCI), Applications of BCI (Neuroprosthetics, communication systems, gaming, rehabilitation)	

	B	Basic Neurophysiology (Neurons, brain activity, brain regions)	CO1, CO2
	C	Types of BCIs: Non-invasive vs. Invasive, Active vs. Passive. History and evolution of BCI technologies	
	Unit 2	**Brain Signal Acquisition**	
	A	Electroencephalography (EEG) – Principles, Electrode placements, and Types.	CO3
	B	Electrocorticography (ECoG) and Magnetoencephalography (MEG)	
	C	Signal Acquisition Hardware: EEG cap, amplifiers, and devices. Signal quality, Noise, and Artifacts in BCI data	
	Unit 3	**Signal Processing for BCI**	
	A	Preprocessing: Filtering (Bandpass, Notch filters), Artifact Removal (Eye blink, Muscle activity). Feature Extraction: Time-domain, Frequency-domain, and Time-Frequency domain features	CO1, CO3
	B	Signal Classification: Machine learning algorithms for BCI (SVM, k-NN, Random Forest)	
	C	Overview of Common Signal Processing Tools and Software (MATLAB, Python libraries)	
	Unit 4	**Ethical, Legal, and Societal Impacts of BCIs**	
	A	Privacy concerns and data security in BCI systems Ethical implications of invasive BCI systems.	CO5
	B	Accessibility and inclusion issues for people with disabilities.	
	C	Legal and regulatory challenges in BCI deployment	
	Unit 5	**Recent Trends and Future Directions in BCI**	
	A	Advances in Non-invasive BCI technologies	CO5, CO6
	B	BCI and Artificial Intelligence (AI) integration	
	C	Brain-to-Brain communication and future possibilities	
	Mode of examination	Theory	
	Weightage Distribution	CA: 25% / ESE: 75%	
	Text book/s*	1. Niedermeyer, E., & da Silva, F. L. (2004). Electroencephalography: Basic Principles, Clinical Applications, and Related Fields. Lippincott Williams & Wilkins. 2. He, H., & Wu, D. (2017). Brain-Computer Interface: Principles and Applications. Springer.	
	Other References	1. Lebedev, M. A., & Nicolelis, M. A. (2006). Brain–machine interfaces: past, present and future. Trends in Neurosciences, 29(9), 536-546.	

		2. Lebedev, M. A., & Nicolelis, M. A. (2006). Brain–machine interfaces: past, present and future. Trends in Neurosciences, 29(9), 536-546.	

PO and PSO mapping with level of strength for Course: CSHC3102- Brain Computer Interface

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
	Engineering Knowledge	Problem Analysis	Design/ Development of Solutions	Conduct Investigations of Complex Problems	Engineering Tool Usage	The Engineer and The World	Ethics	Individual and Collaborative Team work	Communication	Project Management and Finance	Life-Long Learning	Familiarity and practical proficiency	Understand, analyze and develop	Apply standard Software
CO1	3	-	-	-	-	2	-	-	-	-	3	3	3	-
CO2	2	-	-	-	3	2	-	2	-	-	3	3	3	-
CO3	3	3	3	-	3	2	-	-	-	-	2	2	3	-
CO4	3	3	3	3	-	2	2	3	-	-	2	-	-	3
CO5	1	3	2	-	2	2	2	-	-	-	1	-	3	-
CO6	3	3	3	3	3	3	3	3	3	2	3	-	-	3

Average of non-zeros entry in following table (should be auto calculated).

Course Code/ Name	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
	Engineering Knowledge	Problem Analysis	Design/Development of Solutions	Conduct Investigations of Complex Problems	Engineering Tool Usage	The Engineer and The World	Ethics	Individual and Collaborative Team work	Communication	Project Management and Finance	Life-Long Learning	Familiarity and practical proficiency	Understand, analyze and develop	Apply standard Software
CSHC3102/ Brain Computer Interface	3	3.00	2.33	3.00	2.75	2.17	2.33	2.67	3.00	2.00	2.33	2.67	3.00	3.00

Strength of Correlation:

1. Addressed to ***Slight (Low=1) extent***

2. Addressed to ***Moderate (Medium=2) extent***

3. Addressed to ***Substantial (High=3) extent***

School: SSCSE		**Batch: 2025-2029**	
Program: B.Tech		**Current Academic Year: 2025-26**	
Branch: CSE(AR/VR)		**Semester: VI**	
1	**Course Code**	**CSHC3502**	Course Name: **Brain Computer Interface Lab**
2	**Course Title**	**Brain Computer Interface Lab**	
3	**Credits**	**1**	
4	**Contact Hours (L-T-P)**	**0-0-2**	
	Course Status	**Specialization Elective**	
5	Course Objective	1. The practical course aims to provide hands-on experience in designing, implementing, and analyzing Brain-Computer Interface (BCI) systems. 2. 2. Students will work with various types of neural signals, signal processing techniques, and tools used in real-world BCI applications.	
6	Course Outcomes	After the completion of this course students will be able to: **CO1**: Identify and utilize common hardware and software tools used in BCI research such as OpenBCI, NeuroSky, MATLAB, and Python. **CO2**: Acquire EEG signals using appropriate BCI devices and perform preprocessing techniques such as filtering and artifact removal. **CO3**: Extract relevant time-domain and frequency-domain features from EEG signals for subsequent classification. **CO4**: Implement and apply machine learning algorithms (e.g., SVM, k-NN) for classifying EEG signals into cognitive or motor states. **CO5**: Design and implement a basic BCI application that converts brain signals into control commands for a practical task. **CO6**: Create Real Time application BCI system.	
7	Course Description	This practical syllabus is designed to provide a comprehensive foundation in the design, implementation, and analysis of Brain-Computer Interface systems while preparing students for real-world challenges in the field of neuroengineering and assistive technologies.	
8	Outline syllabus		CO Mapping
	Unit 1	**Introduction to BCI Tools and Software**	**CO1**
		Understand the tools used in BCI research (e.g., OpenBCI, NeuroSky) and familiarize with software (e.g., MATLAB, Python).	
	Unit 2	**Signal Acquisition and Preprocessing of EEG Signals.**	**CO2**
		Learn how to acquire EEG signals and preprocess them for further analysis.	
	Unit 3	**Feature Extraction from EEG Signals**	**CO3**

		Extract features from EEG signals for classification purposes.			
	Unit 4	**Signal Classification using Machine Learning Algorithms.**			**CO4**
		Apply machine learning models to classify EEG signals into different cognitive states.			
	Unit 5	**Design and Implementation of a Simple BCI System.**			**CO5, CO6**
		Build a simple BCI system that translates brain signals into commands (e.g., moving a cursor).			
	Mode of examination	Practical			
	Weightage Distribution	CA	CE (Viva)	ESE	
		30%	30%	40%	
	Text book/s*	1. Niedermeyer, E., & da Silva, F. L. (2004). Electroencephalography: Basic Principles, Clinical Applications, and Related Fields. Lippincott Williams & Wilkins. 2. Lebedev, M. A., & Nicolelis, M. A. L. (2006). Brain–computer interfaces: past, present, and future. *Trends in Neurosciences*, 29(9), 536-546			
	Other References	1.He, H., & Wu, D. (2019). Brain-Computer Interface: Principles and Applications. Springer. 2. He, H., Wu, D., & Zhang, Z. (2020). EEG Signal Processing and Classification. Springer.			

PO and PSO mapping with level of strength for Course: CSHC3502 Brain Computer Interface Lab

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO1**	**PSO2**	**PSO3**
	Engineering Knowledge	**Problem Analysis**	**Design/ Development of Solutions**	**Conduct Investigations of Complex Problems**	**Engineering Tool Usage**	**The Engineer and The World**	**Ethics**	**Individual and Collaborative Team work**	**Communication**	**Project Management and Finance**	**Life-Long Learning**	**Familiarity and practical proficiency**	**Understand, analyze and develop**	**Apply standard Software**
CO1	3	-	-	-	-	2	-	-	-	-	3	3	3	-
CO2	2	-	-	-	3	2	-	2	-	-	3	3	3	-
CO3	3	3	3	-	1	2	-	-	-	-	2	2	3	-
CO4	1	3	1	3	-	2	2	3	-	-	2	-	-	3
CO5	2	3	2	-	2	2	2	-	-	-	1	-	3	-
CO6	1	3	3	3	3	3	3	3	3	2	3	-	-	3

Average of non-zeros entry in following table (should be auto calculated).

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO1**	**PSO2**	**PSO3**

School: SSCSE		**Batch: 2023-2027**		
Program: B.Tech.		**Current Academic Year: 2025-26**		
Branch: CSE		**Semester: VI**		
1	Course Code	**CSCR4205** Course Name:**ML Ops**		
2	Course Title	**ML Ops**		
3	Credits	3		
4	Contact Hours (L-T-P)	3	0	0
	Course Status	Core		
5	Course Objective	To provide the students with an understanding of MLOps (Machine Learning Operations) and its role in automating the AI product lifecycle. It covers key MLOps practices such as model development, versioning, CI/CD, containerization, monitoring, and scaling. The course also introduces industry-standard tools for managing ML workflows efficiently in production environments.		
6	Course Outcomes	CO1: To understand MLOps concepts, the AI product lifecycle, and industry best practices. CO2: To implement model development workflows, including versioning and tracking. CO3: To automate machine learning workflows using CI/CD pipelines. CO4: To investigate strategies for Containerization, Deployment, & Monitoring for ML models. CO5: To Optimize ML Model Performance & Scaling using Kubernetes, Kubeflow, and TensorRT, quantization, and pruning techniques. CO6: Evaluate trade-offs in various design decisions regarding performance, compliance, security, and scalability.		
7	Course Description	This course introduces students to MLOps, focusing on integrating ML models into production systems with automation, monitoring, and security. Students will learn about model development, CI/CD pipelines, containerization, deployment strategies, and observability. The course provides hands-on experience with industry-standard tools such as MLflow, Kubernetes, Docker, and cloud-based ML platforms.		
8	Outline syllabus			CO Mapping
	Unit 1	**Introduction to MLOps & AI Product Lifecycle**		
	A	Overview of MLOps: Definition, importance, and key benefits		CO1
	B	AI Product Development: Data-centric AI and responsible AI practices, ML Infrastructure, Open-source end-to-end MLOps Deployments		CO1,
	C	Case Study: Reverse Engineering a Monolithic Web Application.		CO1
	Unit 2	**Model Development, Versioning, & Tracking**		
	A	Data Preparation & Feature Engineering: Handling data and model drift		CO2
	B	Model Training & Evaluation: Experiment tracking with MLflow, DVC		CO2, CO4
	C	Model Versioning & Registries: Managing models efficiently		CO2, CO3
	Unit 3	**Automation & Continuous Integration/Deployment (CI/CD)**		
	A	MLOps Maturity Levels (0-2): Transitioning from manual to automated workflows (CO3)		CO3
	B	CI/CD for Machine Learning: Automating model training and deployment.		CO3

	C	Model Packaging & Dependency Management: Using Docker, Conda, and Virtualenv.	CO3, CO4
	Unit 4	**Containerization, Deployment, & Monitoring**	
	A	Containerization with Docker & Kubernetes: Managing scalable ML models	CO4
	B	Model Deployment Strategies: A/B testing, Blue-Green, and Canary deployments	CO4, CO5
	C	Cloud vs. On-Prem Deployment: AWS SageMaker, GCP AI Platform, Azure ML	CO4, CO5
	Unit 5	**Security, Compliance, & Scaling**	
	A	Observability & Logging: Implementing real-time monitoring and alerting.	CO5
	B	Optimizing ML Model Performance & Scaling: Using Kubernetes, Kubeflow, and TensorRT, quantization, and pruning.	CO5
	C	MLOps Security Risks: Adversarial attacks, vulnerabilities, and risk mitigation.	CO5, CO6
	Mode of examination	Lab	
	Weightage Distribution	CA: 25% / ETE: 75%	
	Text book/s*	1. Noah Gift, Practical MLOps: Operationalizing Machine Learning Models, O'Reilly. 2. Emmanuel Raj, Engineering MLOps, Packt Publishing.Sam Newman, Building Microservices: Designing Fine-Grained Systems, O'Reilly Media, 2021. 3. Yaron Haviv, Implementing Mlops in the Enterprise: A Production-First Approach, O'Reilly Media, 2024.	
	Other References	1. Andrew Ng, Machine Learning Yearning, self-published. 2. Chip Huyen, Designing Machine Learning Systems, O'Reilly. 3. Official Documentation: Kubernetes, MLflow, Docker, TensorFlow Extended (TFX).	

CO and PO Mapping

S. No.	Course Outcome	Program Outcomes (PO) & Program Specific Outcomes (PSO)
1.	CO1: To understand MLOps concepts, the AI product lifecycle, and industry best practices.	PO1, PO2, PO3, PO4, PO5, PO10, PO11, PSO1, PSO2, PSO3
2.	CO2: To implement model development workflows, including versioning and tracking.	PO1, PO2, PO3, PO4, PO5, PO10, PO11, PSO1, PSO2, PSO3
3.	CO3: To automate machine learning workflows using CI/CD pipelines.	PO1, PO2, PO3, PO4, PO5, PO9, PO10, PO11, PSO1, PSO2, PSO3
4.	CO4: To investigate strategies for Containerization, Deployment, & Monitoring for ML models.	PO1, PO2, PO3, PO4, PO5, PO10, PO11, PSO1, PSO2, PSO3

5.	CO5: To Optimize ML Model Performance & Scaling using Kubernetes, Kubeflow, and TensorRT, quantization, and pruning techniques.	PO1, PO2, PO3, PO4, PO5, PO9, PO10, PO11, PSO1, PSO2, PSO3
6.	CO6: Evaluate trade-offs in various design decisions regarding performance, compliance, security, and scalability.	PO1, PO2, PO3, PO4, PO5, PO8, PO9, PO10, PO11, PSO1, PSO2, PSO3

PO and PSO mapping with level of strength for MLOps (Course Code CSCR4205)

	COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO1**	**PSO2**	**PSO3**
CSCR4205 MLOps	**CO1**	3	3	2	3	2	-	-	-	-	2	1	3	3	2
	CO2	3	3	3	3	3	-	-	-	-	2	2	3	3	3
	CO3	3	3	3	3	3	-	-	-	2	2	2	3	3	3
	CO4	3	3	3	3	3	-	-	-	-	2	2	3	3	3
	CO5	3	3	3	3	3	-	-	-	2	3	2	3	3	3
	CO6	2	2	3	3	3	-	-	2	3	3	3	3	3	3

Average of non-zeros entry in following table (should be auto calculated).

Course Code	**Course Name**	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO 1**	**PSO 2**	**PSO 3**
CSCR 4205	**MLOps**	2.83	2.83	2.83	3.00	2.83	0.00	0.00	0.33	1.17	2.33	2.00	3.00	3.00	2.83

Strength of Correlation: 28.98

1. Addressed to***Slight (Low=1)extent*** ***2.*** Addressed to***Moderate (Medium=2) extent***
3. Addressed to***Substantial (High=3) extent***

School:		**Sharda School of Computing Science & Engineering**		
Department		**Department of Computer Science & Engineering**		
Program:		**B.Tech.**		
Branch:		**CSE**		
1	Course Code	CSCR4201		
2	Course Title	Wireless Networks & Mobile Computing: Principles and Applications		
3	Credits	3		
4	Contact Hours (L-T-P)	3	0	0
	Course Status	Core /Elective/Open Elective		
5	Course Objective	1. Provide fundamental knowledge about Wireless networks and mobile computing systems. 2. Understand the concepts of protocol stack and standards, and analyze the network layer solutions for different Wireless networks, and make student aware of 4G and advanced Services. 3. Impart knowledge of mobile computing systems and techniques 4. Compare the performances of different mobile and wireless communication protocols.		
6	Course Outcomes	CO1: To analyze existing software systems to understand their architecture, design choices, and scalability. CO2. To identify and describe the infrastructure and requirements of Mobile IP and Mobile IPv6 and principles of TCP CO3.To illustrate the issues and solutions of various layers of mobile networks, namely MAC layer, Network Layer & Transport Layer and identify and describe the structure of current 4G cellular networks. CO4: To study Multicarrier Modulation, Smart antenna techniques, OFDM-MIMO systems and synthesize the concepts of IEEE802.11, Bluetooth and HYPERLAN CO5: To understand the concept of mobile IP & various Routing Protocols and synthesize the concepts of Mobile Transport Layer & WAP. CO6: To compare the performances of different mobile and wireless communication protocols.		
7	Course Description	The course describes concepts, technology and applications of wireless networking used in current and next-generation wireless networks. In addition, the course addresses the fundamentals of wireless communications and provides an overview of existing and emerging wireless communication networks. This course will also cover various topics of mobile computing, networking, and systems, that includes applications of smart phones, cellular networks, embedded sensor systems, localization systems, energy efficiency of mobile devices, wearable and vehicular mobile systems, mobile security etc.		

8	Outline syllabus		CO Mapping
	Unit 1	**Introduction to Wireless Networks and Mobile Computing**	
	A	Introduction-WLAN technologies: Wireless transmission, Signals , Antennas , Propagation, Infrared, UHF narrowband, spread spectrum - IEEE802.11: System architecture, protocol architecture,	CO1, CO2
	B	Spread spectrum, MAC, SDMA , FDMA , TDMA , CDMA , Cellular Wireless Networks	CO1, CO2
	C	IEEE802.16-WIMAX: Physical layer, MAC, Spectrum allocation for WIMAX	CO1, CO2
	Unit 2	**Mobile Network and Transport Layer**	
	A	Introduction - Mobile IP: IP packet delivery, Agent discovery, tunneling and encapsulation.	CO2, CO4
	B	IPV6-Network layer in the internet Mobile IP session initiation protocol, Mobile ad-hoc network: Routing Destination Sequence distance vector, Dynamic source routing.	CO2, CO4
	C	TCP enhancements for wireless protocols - Traditional TCP: Congestion control, fast retransmit/fast recovery, Implications of mobility, Classical TCP improvements: Indirect TCP, Snooping TCP, Mobile TCP, Time out freezing	CO2, CO4
	Unit 3	**Wireless Wide are, 4G and Advanced Networks**	
	A	Overview of UTMS Terrestrial Radio access network, 3G-MSC, 3G-SGSN, 3G-GGSN	CO3, CO4
	B	DNS/DHCP-High speed Downlink packet access (HSDPA)- LTE network architecture and protocol.	CO3, CO4
	C	Introduction – 4G vision – 4G features and challenges - Applications of 4G – 4G Technologies, Multicarrier Modulation, Smart antenna techniques, OFDM-MIMO systems,	CO3, CO4
	Unit 4	**Telecommunication Networks**	
	A	Multicarrier Modulation, Smart antenna techniques, OFDM-MIMO systems,	CO4, CO5
	B	GPRS Architecture, GPRS network nodes.	CO4, CO5
	C	Introduction to IEEE 802.11b/g/n, Bluetooth technologies and architecture, Bluetooth technologies and architecture.	CO4, CO5
	Unit 5	**Layers and Protocols for Mobile Networks**	
	A	Mobile IP Goals, Entities, IP packet Delivery Agent Advertisement and Discovery, Registration,	CO5, CO6
	B	Hidden and exposed terminal problems ,Routing protocols classification, DSDV, DSR, AODV ,Security	CO5, CO6
	C	Traditional TCP, Indirect TCP, Snooping TCP, Mobile TCP, WAP: Protocols, Architecture	CO5, CO6

	Mode of examination	Theory/Jury/Practical/Viva		
	Weightage Distribution	CA	ETE	
		25%	75%	
	Text book/s*	1. Jochen Schiller, Mobile Communications, Second Edition, Pearson Education 2012. 2. U. Hansman and L. Merck : Principles of Mobile Computing", 2nd Ed., Springer		
	Other References	1. Erik Dahlman, Stefan Parkvall, Johan Skold and Per Beming, "3G Evolution HSPA and LTE for Mobile Broadband", Second Edition, Academic Press, 2008. 2. Anurag Kumar, D.Manjunath, Joy kuri, "Wireless Networking", First Edition, Elsevier 2011. 3. Simon Haykin , Michael Moher, David Koilpillai, "Modern Wireless Communications", First Edition, Pearson Education 2013 Gaurav Sen, System Design Interview – An Insider's Guide, Self-Published, 2022. 4. A. S. Tanenbaum. : Computer Networks, 4th Ed., Pearson Education. 5. D. Milojicic, F. Douglis. : Mobility Processes, Computers and Agents",Addison Wesley 6. D.B. Lange and M. Oshima : Programming and Deploying Java Mobile Agents with Aglets, Addison Wesley.		

PO and PSO mapping with level of strength for System Design (Course Code: CSCR4201)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PSO1	PSO2	PSO3
	Engineering Knowledge	Problem Analysis	Design/Development of	Conduct Investigation	Engineering Tool Usage	The Engineer	Ethics	Individual and	Communication	Project Management	Life-Long Learning	Familiarity and practical	Understand, analyze and	Apply standard Software
CO1	3	3	2	3	2	-	-	-	-	2	1	3	3	2
CO2	3	3	3	3	3	-	-	-	-	2	2	3	3	3
CO3	3	3	3	3	3	-	-	-	2	2	2	3	3	3
CO4	3	3	3	3	3	-	-	-	-	2	2	3	3	3
CO5	3	3	3	3	3	-	-	-	2	3	2	3	3	3
CO6	2	2	3	3	3	-	-	2	3	3	3	3	3	3

Average of non-zeros entry in following table (should be auto calculated).

Course Code	Course Name	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
		Engineering	Problem Analysis	Design/Development	Conduct Investigati	Engineering Tool	The Engineer	Ethics	Individual and	Communication	Project Managem	Life-Long Learning	Familiarity and	Understand, analyze	Apply standard
CSCR4201	Wireless Networks & Mobile Computing: Principles and Applications	2.83	2.83	2.83	3.00	2.83	0.00	0.00	0.33	1.17	2.33	2.00	3.00	3.00	2.83

Strength of Correlation: 29

1. Addressed to ***Slight (Low=1) extent*** ***2.*** Addressed to ***Moderate (Medium=2) extent***
3. Addressed to ***Substantial (High=3) extent***

Weekly Topic List												
Week	Date	CODE	(ARP 208/210)		ARP 208		SKP 4004		ARP 320		ARP 306	
		SEMESTER	2nd/4th		4th Sem		4th		4th Sem		6th Sem	
		SCHOOL	ARP 208 - SSCSE - B.Tech CSE/IT (4th sem)		ARP 208 - SSBS - BBA/BCOM (4th sem)		SKP 4004 - SHSS/SBSR/SMFE/SAHS - (4th Sem)		ARP 320 - SSCSE - MCA/MSC		SSCSE - BTECH CSE/IT- 6TH SEM	
		DOMAIN	QA/LR (1 class/week)	VA/SS (2 classes/week)	QA/LR (1 class/week)	VA/SS (1 class/week)	LR (1 class/week)	VA/SS (2 classes/week)	QA/LR (2 classes/week)	VA/SS (1 class/week)	QA/LR (2 classes/week)	VA/SS (1 class/week)
Week 1	5 Jan - 9 Jan	**All Working Days**	Number Puzzles	1. The Big 5 personality Traits 2. SWOT ANALYSIS	Number Puzzles	The Big 5 personality Traits and SWOT ANALYSIS	Number Series	1. Understanding Personality – Meaning, components, and importance of personality 2. Individual Differences & Personality Types – Nature vs nurture, personality styles, temperament	1. Problem on Ages 2. Averages	Personal Branding	1. Problem on Ages 2. Profit & Loss	PROFESSIONAL GROOMING - Dress for Success
Week 2	12 Jan - 16 Jan	**Wednesday - 14 Jan - Holiday (Makar Sankaranti)**	Mathematical Operations	1. Basics of Communication - Developing Effective Communication Skills 2. 7 C's of Communication	Mathematical Operations	7 C's of Communication and Business Communication - Email Etiquettes & Telephone Etiquette - Non Verbal Communication - Developing Sensory Acuity - VAKOG	Clocks	1. Personality Development & Personal Transformation – Mindset shift, habits, grooming, attitude 2. Creating a Positive Impression: The 3 V's of Impact – Verbal, Vocal & Visual communication	1. Allegation & Mixtures 2. Ratio & Proportions	Uploading & Curating Resumes in Job Portals, getting Your Resumes Noticed	1. Simple Interest 2. Compound Interest	Profiling a Job to Meet Your Most Dominant Strengths/ SWOT
Week 3	19 Jan - 23 Jan	**All Working Days**	Blood Relations (All Types)	1. Business Communication - Email Etiquettes & Telephone Etiquette 2. Non Verbal Communication - Developing Sensory Acuity - VAKOG	Quadratic Equations	Sentence Improvement - I (TCS/WIPRO/TECH MAHINDRA/ACCENTURE)	Deductive Reasoning	1. Self-Awareness & Self-Assessment – Knowing strengths, weaknesses, values 2. Building Self-Confidence & Self-Esteem – Overcoming fear, self-doubt, confidence techniques	1. Probability 2. Probability Practice	Writing Cover Letters	1. Puzzles (Floor based) 2. Puzzles (box based)	How to Upload Resumes in Job Portals with Cover Letters
Week 4	27 jan - 30 Jan	**All Working Days**	Averages	1. Sentence Improvement - I (TCS/WIPRO/TECH MAHINDRA/ACCENTURE) 2. Grammar Rule Series	Averages	Self Esteem & Self Confidence - connected to Professional Grooming	Blood Relations	1. Behavioural Skills & Professional Conduct – Workplace etiquette, body language, emotional control 2. Interpersonal Skills & Relationship Building – Empathy, listening skills, collaboration	1. Permutation & Combination 2. Permutation & Combination Practice	Conflict and Relationship Management	1. Visual Flow Chart 2. Game based puzzles	SPOTTING THE ERRORS - Double and triple errors -II
Week 5	2 Feb - 5 Feb	**Thursday - 5 Feb (Classes will suspend after 2 pm due to Chorus)**	**Doubt Session/Revision**	1. Self Esteem & Self Confidence 2. Sentence Correction - I (TCS/WIPRO/TECH MAHINDRA/ACCENTURE)	**Doubt Session/Revision**	Paragraph Completion and PARAGRAPH SUMMARY - I	Direction & Sense	1. Assertiveness Skills & Avoiding Arguments – Saying no professionally, standing your ground 2. Art of Constructive Feedback & Criticism – Giving and receiving feedback positively	**1. Doubt Session/Revision 2. Test 1**	Sentence Correction	**1. Revision/Doubt Session 2. Test 1**	LinkedIn Profile Optimization
Week 6	9 Feb - 13 Feb	**All Working Days**	Quadratic Equations	1. Paragraph Completion 2. Professional Grooming	Blood Relations (All Types)	Mastering the Art of Giving & Seeking Feedback - e.g. SANDWICH MODEL	Puzzles (Floor | Week | Box Based)	1. Personal Effectiveness Grid – Urgency vs importance, prioritization skills 2. Time & Energy Management – Productivity tools, work-life balance	1. Simple Interest 2. Compound Interest	HR Sensitization (Role Clarity | KRA | KPI | Understanding JD)	1. Boats & Streams 2. Non verbal Series	CRITICAL REASONING - Moderate to Advanced
Week 7	16 Feb - 20 Feb	**Assessment 1 - (19 Feb - 20 Feb)**	Cubes & Dices	1. The Art of Rapport Building & Empathetic Listening 2. PARAGRAPH SUMMARY - I	Cubes & Dices	Resumes and Modern/Internship Job Hunt Strategies - Naukri.com/LinkedIn (ATS OPTIMIZED	Cubes & Dices	1. Assessing Strengths & Limitations – SWOT analysis, self-review techniques 2. Action Planning for Continuous Learning (4M Model) – Mindset, Method, Mentors, Measurement	1. Data Sufficiency (Logical Reasoning) 2. Odd One Out | Cause & Effect	Antonyms and Synonyms	1. Permutation & Combination 2. Permutation & Combination Practice	Networking- Professional Relationship Building
Week 8	23 Feb - 27 Feb	**Assessment 1 - (23 Feb)**	Direction & Sense	1. Mastering the Art of Giving & Seeking Feedback - e.g. SANDWICH MODEL 2. Test	Direction & Sense	PARAGRAPH JUMBLES - I (TCS/WIPRO/TECH MAHINDRA/ACCENTURE)	Visual Flowchart	1. Fundamentals of English Usage – Grammar basics, sentence structure 2. Common Errors in Communicative English – Spoken & written mistakes to avoid	1. Sitting Arrangement (Linear & Circular) 2. Sitting Arrangement (Square & Rectangle)	Idioms and Phrases	1. Probability 2 Probability Practice	Parajumbles- Advance level

Week 9	2 Mar - 6 Mar	**Wednesday - 4 Mar (Holiday - Holi)**	Partnerships	1. VMOSA (Vision, Mission, Values and Ethics) 2. WriteX - Emails/Letters/Applications	Partnerships	VMOSA (Vision, Mission, Values and Ethics)	Venn Diagram	1. Vocabulary Building & Word Power – Synonyms, professional vocabulary, usage 2. Reading Comprehension Skills – Techniques for understanding passages	1. Puzzles (Floor Based & Box Based) 2. Puzzles (Date Based & Week Based)	Negotiation Skills	1. Time and Work 2. Time and Work Practice	GROUP DISCUSSION- Case Study/Abstract
Week 10	9 Mar - 13 Mar	**All Working Days**	AP & GP	1. Modern/Internship Job Hunt Strategies - Naukri.com/LinkedIn (ATS OPTIMIZED) 2. WriteX - Essay	AP & GP	WriteX - Emails/Letters/Applications	Seating Arrangement (Linear & Circular)	1. Reading Comprehension Practice – Application-based exercises 2. Verbal Reasoning & Logical Ability – Critical thinking, verbal logic, reasoning skills	1. Direction & Sense 2. Odd One out	Spotting the Errors	1. Puzzle (Date Based) 2. Puzzle (Week Based)	Cloze Test
Week 11	16 Mar - 20 Mar	**Thursday - 19 Mar (Holiday - Ramzan Id)**	Inequalities (Reasoning)	1. Resumes 2. PARAGRAPH JUMBLES - I (TCS/WIPRO/TECH MAHINDRA/ACCENTURE)	Inequalities (Reasoning)	READING COMPREHENSION - I (Short Passages & Long Passages) - TCS/WIPRO/TECH MAHINDRA/ACCENTURE	Seating Arrangement (Rectangle & Square)	1. Applied Communication Skills (Domain-Specific) – Corporate, academic, customer-facing communication 2. Email Writing & Business Communication – Professional emails, tone, clarity	**1. Revision/Doubt Session 2. Test 2**	**Test**	**1. Revision/Doubt Session 2. Test 2**	Personal Interviews and Interview panel Expectations
Week 12	23 Mar - 27 Mar	**All Working Days**	Order & Ranking	1. Interview Skills & PI Practice 2. READING COMPREHENSION - I (Short Passages & Long Passages) - TCS/WIPRO/TECH MAHINDRA/ACCENTURE	Order & Ranking	Interview Skills & PI Practice	Critical Reasoning (Statement & Conclusion | Statement & Assumption)	1. Resume Building & Personal Branding – ATS-friendly resumes, profile positioning 2. Cover Letter Writing – Customization, structure, impact	1. Statement & Conclusion 2. Strong & Weak Arguments	Reading Comprehension	1. Mensuration 2. Deductive Reasoning	HR Sensitization (Role Clarity | KRA | KPI | Understanding JD)
Week 13	30 Mar - 2 Apr	**Assessment 2 - (2 Apr)**	Syllogism (Either or/Possibility)	1. Group Discussion Practice 2. Cloze Test	Syllogism (Either or/Possibility)	Group Discussion Practice	Critical Reasoning (Sause & Effect | Statement & Arguments)	1. Presentation Skills – Foundations – Structure, confidence, content flow 2. Presentation Skills – Advanced – Storytelling, visuals, handling questions	1. Selection Based on Given Conditions) 2. Analogies (All types)	Cloze Test	1. Statement & Arguments 2. Selection Based on given conditions	Conflict Resolution- Case Studies
Week 14	6 Apr - 10 Apr	**Assessment 2 - (6 Apr - 8 Apr)**	Analogies (All Types)	1. Critical Reasoning - I (Argument Structure, Patters) 2. Soft Skill Practice	Analogies (All Types)	Cloze Test	Data Sufficiency (Reasoning)	1. Group Discussion Skills – Participation, leadership, clarity of thought 2. Personal Interview Skills – HR, technical & behavioral interviews	**Full Length Test**	**Full Length Test**	**Full Length Test**	Full Length Test
Week 15	13 Apr - 16 Apr	**Tuesday - 14 Apr (Holiday - Dr. Ambedkar Jayanti)**	**Doubt Session/Revision**	1. Critical Reasoning - II (Strategy for Elimination) 2. Test	**Doubt Session/Revision**	Critical Reasoning - I (Argument Structure, Patters)	Mirror & Water Image	1. GD & PI Practice Sessions – Mock sessions, feedback & improvement 2. HR Sensitisation & Organisational Understanding – Role clarity, JD, KRA, KPI, structure, functions & departments	**Doubt Session/ Revision**	**Doubt Session/ Revision**	**Doubt Session/ Revision**	Doubt Session/ Revision